
Candidate-Conditioned Machine-Learning Coincidence Scoring for Spectrally Broadened Signals in Dual- Site LOFAR Data

Methods, validation, results, and research outlook

Zainab Rehman Naz

Breakthrough Listen Research Internship 2026 – University of Galway / Berkeley SETI

Project repository: github.com/zainab-rh/lofts-broadened-coincidence



Contents

1	Executive summary	2
2	Scientific and operational context	2
3	Inherited foundation and contributions of this work	3
4	Stage-3 broadened-signal study	4
5	Candidate-conditioned Stage 4	5
6	Controlled Test A: downstream coincidence performance	7
7	Real LOFTS0050 Ireland–Sweden pilot	9
8	Blind BLISS recovery of a broadened injection on real backgrounds	11
9	Interpretation of scores and terminology	12
10	Next research phase: multi-event Test B	13
11	Conclusion	13

1 Executive summary

The project addresses a specific cross-station question: after BLISS has reported a candidate at either the Ireland or Sweden LOFAR station, are the two barycentric observations mutually consistent with the same physical event? The two sites share the sky signal but not the same local interference or noise environment. A genuine shared event should therefore be compatible in physical frequency and signed drift, while its signal-to-noise ratio may differ between stations.

The final system separates search from coincidence scoring. BLISS remains the upstream single-station detector. It searches each station independently and reports candidate frequency, signed drift, spectral width, and a search statistic. The downstream pipeline forms the union of the Ireland and Sweden candidate catalogues, projects every candidate into the corresponding physical location in the other station, prepares both views using the candidate measurements, and then assigns a two-station consistency score. A detection at only one station is therefore retained rather than discarded.

Main outcomes

- The inherited Stage-3 model was evaluated and fine-tuned on broadened signals. The study showed that raw-view coincidence performance degrades as spectral width increases, motivating candidate-informed representation rather than further reliance on an unchanged raw cutout.
- Stage 4 introduced robust candidate-conditioned preprocessing: signed de-chirping, recentering, and width integration based on the BLISS-reported candidate parameters.
- Stage 4 reused the inherited shared encoder through transfer learning and added a symmetric pair representation and learned MLP head, so station order does not affect the score.
- On the frozen controlled Test-A population of 10,800 simulated matched and mismatched candidate pairs spanning 10–100 Hz, Stage 4 achieved $\text{AUC}_{\text{ROC}} = 0.993$ compared with 0.818 for Stage 3. True-match recovery increased from 44.7% to 96.3% while the false-positive rate remained approximately 3%.
- The real LOFTS0050 Ireland–Sweden pilot processed independent BLISS catalogues through physical-coordinate union, paired extraction, candidate-conditioned scoring, and same-observation shifted controls.
- A paired blind-injection recovery validation inserted a 30 Hz broadened event into separate copies of the real station backgrounds. Independent BLISS searches recovered the event at both stations with frequency and signed-drift estimates consistent with the injected track.

2 Scientific and operational context

2.1 Why a dual-site coincidence comparison is useful

The Ireland and Sweden stations observe the same astronomical field from geographically separated sites. A sky-origin signal can therefore be present at both stations while local radio-frequency interference is less likely to reproduce the same physical track at both sites. The useful coincidence variables are the physical quantities that describe the track - barycentric reference frequency, signed drift, and width - rather than channel indices, because independently processed station products can have different frequency grids.

A shared event is expected to be compatible in barycentric frequency and signed drift at a common reference epoch. Its amplitude and S/N are not required to match. This matters operationally: a real signal may be detected by BLISS in one station but remain below the single-station search threshold in

the other. The union-based design preserves that case.

2.2 Search and coincidence are separate stages

BLISS and Stage 4 have deliberately separate roles. BLISS performs the blind single-station search. The downstream coincidence code begins from the union of the two BLISS catalogues, extracts the corresponding physical region from both stations, applies candidate-conditioned preparation, and assigns the cross-station score. Stage 4 therefore does not replace BLISS and does not require an independent BLISS detection at both stations.

Operational rule

Search first, compare second. A candidate reported by either station remains in the union. The other station is sampled at the corresponding barycentric physical coordinates even when BLISS did not independently report a hit there.

3 Inherited foundation and contributions of this work

This project builds directly on the Stage-3 comparator developed in the previous internship. Table 1 distinguishes that inherited foundation from the extensions developed in this work.

Area	Inherited starting point	Contribution developed in this work
Representation	U-Net-like shared encoder with 32/64/128 convolutional blocks, 128-channel bottleneck, adaptive pooling, and a 512-dimensional L2-normalised projection.	Evaluated and fine-tuned the inherited representation for broadened signals; diagnosed the width-dependent failure of the raw-view comparison.
Training objective	Decoder reconstruction objective plus projection-space contrastive objective.	Reused the inherited encoder through transfer learning and added a Stage-4 BCE objective with a weighted contrastive regulariser.
Inference rule	Stage-3 comparison based on Euclidean distance between the two station embeddings.	Added candidate-conditioned preparation and a learned symmetric pair head.
Candidate handling	Pairwise comparison assumed a nominated region.	Built the physical-coordinate Ireland–Sweden candidate union so one-station BLISS detections remain eligible for cross-station comparison.
Validation	Narrowband and earlier broadened diagnostics.	Built the frozen controlled Test A, the real LOFTS0050 pilot and controls, and the paired blind-injection recovery validation; defined the locked multi-event Test-B design.

Table 1: Inherited components and the main technical contributions of this internship.

3.1 Development path

The stage numbers describe model development, while Test A, the real pilot, and Test B describe validation questions. They should not be interpreted as a single linear runtime pipeline.

Component	Technical role	Question answered
Stage 3	Inherited shared-encoder comparator, evaluated and fine-tuned on broadened cases.	How well does the raw-view representation remain discriminative as signal width increases?
Stage 4	Candidate-conditioned representation plus a station-order-invariant learned pair head.	Can BLISS measurements be used to compare broadened candidates more effectively?
Controlled Test A	Same frozen 10–100 Hz matched/mismatched candidate pairs scored by parallel comparison methods.	Once a candidate is available, how well are matches ranked above mismatches?
Real LOFTS0050 pilot	Independent real BLISS catalogues, physical union, paired extraction, scoring, and shifted controls.	How does the frozen workflow behave in the actual two-station candidate and RFI environment?
Paired blind-injection validation	Known broadened event added to separate real-background copies before independent BLISS searches.	Does the upstream injection–search–recovery interface work with the real station geometry?
Full Test B campaign	Multiple truth-segregated paired injections and deliberate noncoincident cases, with truth revealed after search and inference.	What are BLISS completeness, Stage-4 conditional coincidence performance, and full end-to-end performance?

Table 2: Relationship between model stages and validation experiments.

4 Stage-3 broadened-signal study

4.1 Inherited Stage-3 comparator

The inherited Stage-3 model is a shared-weight, U-Net-style comparator. The same encoder processes the raw Ireland and Sweden time–frequency cutouts independently; in that sense the comparison is Siamese-style. Three convolutional blocks with 32, 64, and 128 channels lead to a 128-channel bottleneck. Adaptive pooling and a projection head produce a 512-dimensional L2-normalised embedding for each station view. During Stage-3 training, the decoder supplies a reconstruction objective and the projection branch supplies the contrastive comparison objective.

At inference, Stage 3 does not compare the two waterfall images by eye and does not first use BLISS width or drift to align them. If the two embeddings are z_A and z_B , the inherited comparison statistic is their Euclidean distance,

$$d_3 = \|z_A - z_B\|_2. \quad (1)$$

Smaller distance means a more similar pair. Evaluation code uses the equivalent score direction in which more match-like pairs rank above mismatches. Thus, the Stage-3 baseline is specifically **raw cutout** → **shared encoder** → **embedding distance**; the candidate-informed alignment introduced later is not part of this baseline.

4.2 Why Stage 4 was needed

The broadened-aware Stage-3 experiments retained the inherited architecture while introducing broadened signal cases. The width sweep in Figure 1 shows the key transition: performance is strong for narrow signals, improves in the accessible broadened regime after fine-tuning, and then approaches random-ranking behaviour as width increases and the raw representation disperses the signal across frequency. The corresponding latent-space diagnostics show the same loss of class separation for difficult broadened cases.

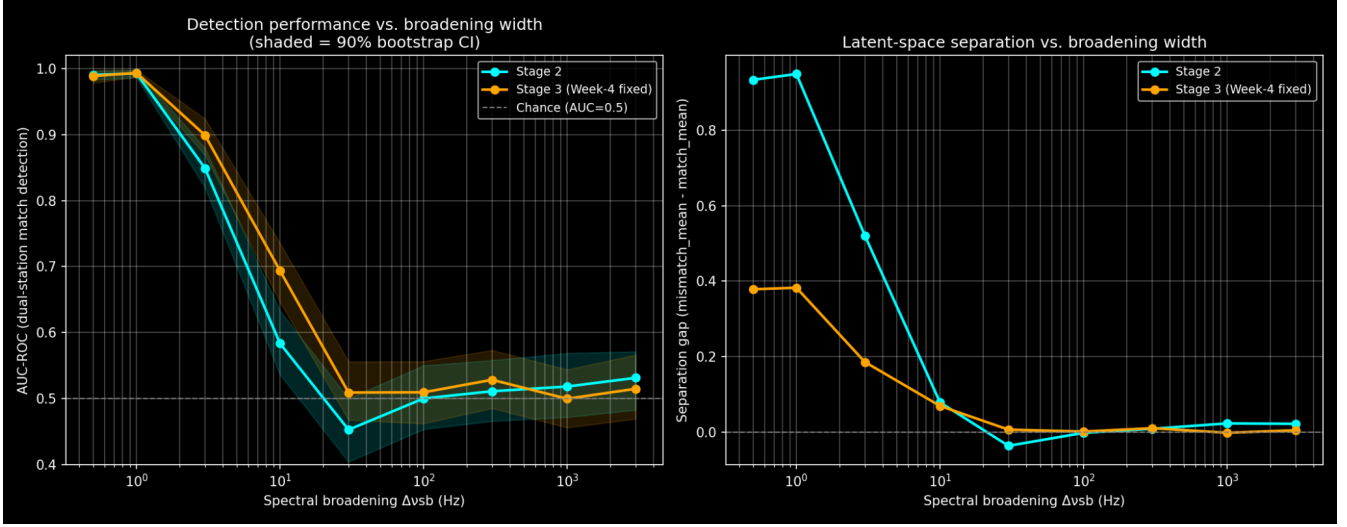


Figure 1: Stage-3 broadened-signal width sweep. Fine-tuning improves the accessible broadened regime, but the raw-view comparison loses discrimination as spectral broadening increases. This is the direct motivation for candidate-conditioned representation.

The width sweep therefore identified the raw representation, rather than the shared encoder itself, as the principal limitation for broader candidates. Stage 4 preserves the learned representation while moving known candidate physics—drift, centre, and width—into the preparation and comparison design.

5 Candidate-conditioned Stage 4

5.1 Physical-coordinate candidate union

BLISS searches Ireland and Sweden independently. Let C_{IRL} and C_{SWE} denote the resulting catalogues. The downstream set is the union

$$C_{\text{union}} = C_{\text{IRL}} \cup C_{\text{SWE}},$$

formed in physical frequency and signed-drift coordinates rather than channel number. Frequency is propagated to a declared common epoch using the signed drift. If a candidate appears only in Sweden, the corresponding physical location is still extracted from Ireland; the reverse applies to Ireland-only detections. This is a design requirement, not a fallback: S/N can differ substantially between the two stations even for a shared sky event.

5.2 Candidate-conditioned preprocessing

For a time–frequency tile $X(t, k)$, the BLISS drift estimate is converted to a row-dependent channel displacement using the signed channel increment δf :

$$\Delta k(t) = -\frac{\dot{f}(t - t_{\text{ref}})}{\delta f}. \quad (2)$$

Retaining the sign of δf is essential because the stored LOFAR frequency axis can descend with channel number.

Each station view is prepared independently:

1. robust local bandpass removal and whitening, while masking the candidate corridor;

2. signed de-chirping according to Eq. 2, with zero-filled and validity-masked edges;
 3. recentering of the straightened track in a fixed-size output tile; and
 4. integration across the BLISS-reported FWHM with a channel-integrated, unit-L2 frequency kernel.
- The primary experiment uses a boxcar kernel.

The purpose is to gather power that is already distributed across the reported width and present both stations in a common candidate-centred frame. The source observations are unchanged.

5.3 Symmetric learned comparison

The prepared views pass through the same inherited shared-weight encoder, using the same network parameters for Ireland and Sweden. For normalised embeddings $z_A, z_B \in \mathbb{R}^{512}$, Stage 4 constructs the symmetric feature vector

$$\left[|z_A - z_B|, z_A \odot z_B, \|z_A - z_B\|_2, z_A^\top z_B, \phi(X_A, X_B) \right], \quad (3)$$

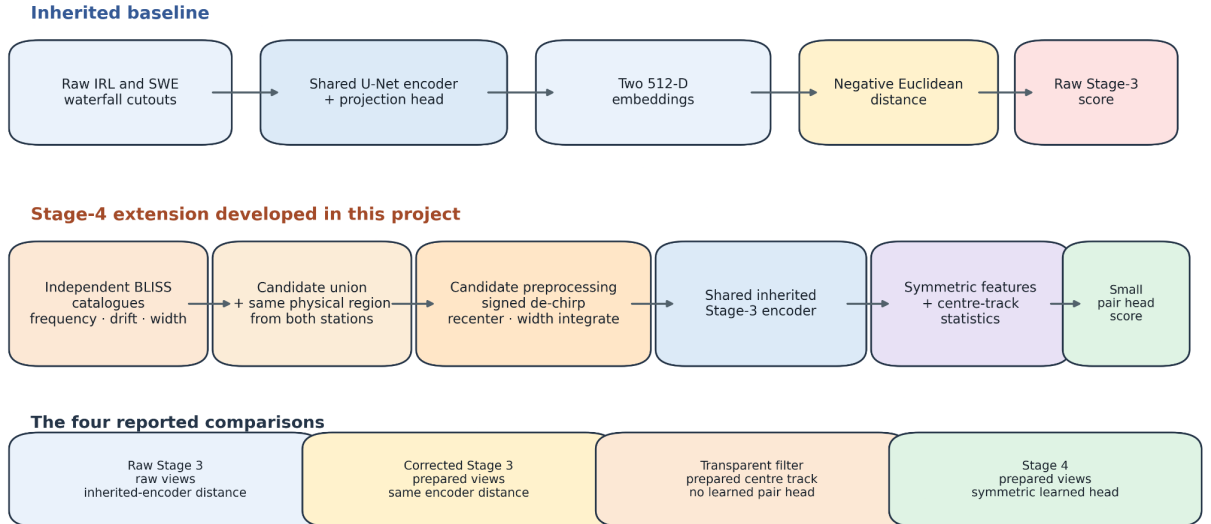
where ϕ contains symmetric central-track summaries and a track correlation. The full vector has 1051 components and is passed through an MLP

$$1051 \rightarrow 256 \rightarrow 64 \rightarrow 1,$$

with layer normalisation, GELU activations, and dropout. A sigmoid maps the final logit to the Stage-4 ranking score. Because every pair feature is symmetric, swapping Ireland and Sweden leaves the score unchanged.

Training uses binary cross-entropy plus a weighted contrastive regulariser. The new pair head is trained first; the upper encoder blocks and projection head are then unfrozen for transfer learning. The inherited decoder is not used for Stage-4 inference.

From the inherited Stage-3 comparator to the candidate-conditioned Stage-4 pipeline



Stage 4 does not replace BLISS: BLISS says where to look; Stage 4 asks whether the two prepared station views are mutually consistent.

Figure 2: Architecture summary. Stage 3 compares raw station representations through Euclidean distance. Stage 4 retains the inherited shared encoder but adds BLISS-informed pairing, signed de-chirping, recentering, width integration, symmetric pair features, and a learned pair head. The lower row identifies the four comparison methods used in the evaluation.

5.4 The four evaluation methods

The four labels reported in the frozen Test-A results are parallel evaluation methods, not successive stages of the deployed pipeline.

Raw Stage 3 is the inherited baseline: the raw Ireland and Sweden cutouts are passed through the shared Stage-3 encoder and compared using embedding distance.

Corrected Stage 3 applies the BLISS-informed candidate preparation but retains the same frozen Stage-3 encoder and embedding-distance comparison. It therefore isolates the effect of changing the input representation without changing the learned comparison.

The transparent filter uses the same candidate-prepared views but scores the prepared centre-track evidence directly, using a model-free statistic based on the weaker of the two station views. It does not use the learned Stage-4 pair head. Its purpose is to quantify how much discrimination is provided by the candidate-conditioned signal processing itself.

Stage 4 uses the same prepared views, passes them through the inherited shared encoder, and combines the resulting representations with symmetric pair features and centre-track summaries in the learned pair head.

Method	Input representation	Scoring rule	What it isolates
Raw Stage 3	Raw paired cutouts	Inherited shared-encoder distance	Baseline performance before candidate-informed preparation.
Corrected Stage 3	Candidate-prepared cutouts	The same frozen Stage-3 encoder and distance	Effect of candidate preparation without changing the learned comparison.
Transparent filter	Candidate-prepared cutouts	Model-free weaker-station central-track statistic	Strength of the candidate-conditioned signal-processing representation without a learned pair head.
Stage 4	Candidate-prepared cutouts	Inherited shared encoder plus symmetric learned pair head	Full candidate-conditioned learned comparison.

Table 3: Exact interpretation of the four methods in the frozen Test-A and real-pilot outputs.

6 Controlled Test A: downstream coincidence performance

6.1 Evaluation design

Test A evaluates the downstream comparison on a frozen detected-conditioned population: 10,800 simulated matched and mismatched candidate pairs spanning 10–100 Hz and three profile families (Lorentzian, Gaussian, and box). Every method is evaluated on the same pairs. The primary metric is threshold-free AUC_{ROC} ; paired differences use 2,000 paired bootstrap replicates with 95% confidence intervals.

For interpretation, $\text{AUC}_{\text{ROC}} = 0.5$ corresponds to random ranking of a match versus a mismatch, while $\text{AUC}_{\text{ROC}} = 1$ corresponds to perfect separation.

Method	AUC _{ROC}	Change from Raw Stage 3
Raw Stage 3	0.81775	reference
Corrected Stage 3	0.83307	+0.015
Transparent filter	0.97470	+0.157
Stage 4	0.99317	+0.175

Table 4: Frozen Test-A AUC values on the same 10,800 simulated candidate pairs.

6.2 Primary AUC result and ablation

The direct Stage-3-to-Stage-4 comparison is AUC_{ROC} 0.81775 $\rightarrow$ 0.99317. The paired improvement is +0.17542 with 95% CI [+0.16790, +0.18328]. The transparent filter already reaches 0.97470, showing that candidate-informed representation explains most of the gain over raw Stage 3. Stage 4 then adds a smaller but resolved learned improvement over the filter: +0.018470, with 95% CI [+0.015609, +0.021387].

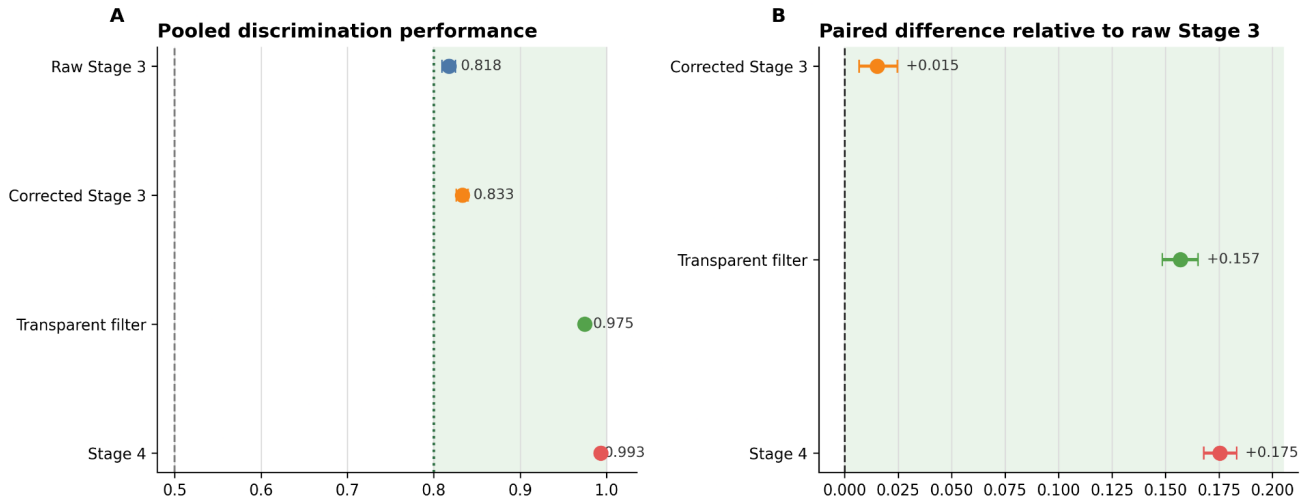


Figure 3: Frozen Test-A primary endpoint. The left panel compares pooled AUC across all four methods; the right panel shows paired AUC improvement relative to Raw Stage 3. The figure makes the ablation result explicit: preprocessing provides the largest gain and the Stage-4 learned head adds an additional improvement.

6.3 Practical operating-point result

At the frozen decision thresholds, Stage 3 obtained precision 0.933, recall 0.447, $F_1 = 0.604$, and false-positive rate 3.20%. Stage 4 obtained precision 0.969, recall 0.963, $F_1 = 0.966$, and false-positive rate 3.13%. In count form, true matches recovered increased from 2,412 of 5,400 to 5,201 of 5,400, while false positives changed only from 173 to 169.

True-match recovery: 44.7% $\rightarrow$ 96.3%

False-positive rate remained approximately 3%.

Operating-point comparison on the detected-conditioned test population

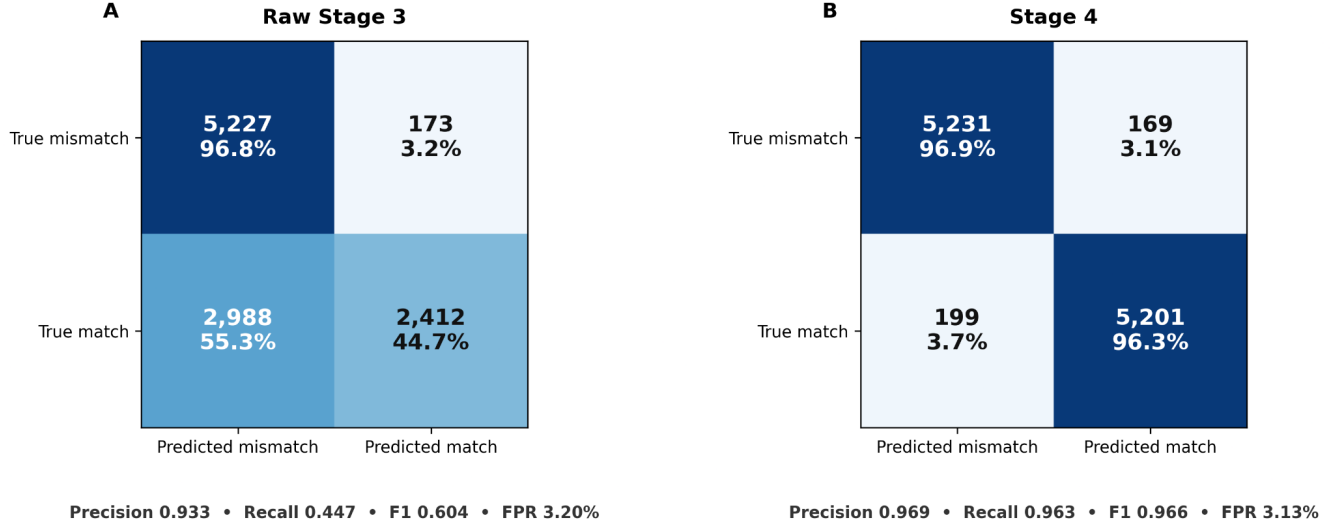


Figure 4: Frozen operating-point comparison. Stage 4 recovers 96.3% of true matches while keeping the mismatch false-positive rate at approximately the same level as Stage 3.

6.4 Width and profile behaviour

The width- and profile-resolved analysis is important for understanding what Stage 4 is solving. In the detected-conditioned population, Stage-4 AUC remains high across the full 10–100 Hz interval and across Lorentzian, Gaussian, and box profiles. The lowest displayed cell is 0.966. This shows that once a usable broadened candidate is available, the candidate-conditioned representation remains highly discriminative throughout the target width range.

The fixed-power population answers a different question. When total signal power is held fixed and spread over progressively more frequency channels, performance declines with width because the local evidence itself becomes weaker. The pooled fixed-power AUC values are correspondingly lower than the detected-conditioned results. This separation is important for end-to-end interpretation: Stage 4 addresses the downstream coincidence comparison, while the upstream probability that BLISS finds a weak broadened signal is measured separately.

7 Real LOFTS0050 Ireland–Sweden pilot

7.1 Input geometry and BLISS candidate population

The real pilot uses simultaneous barycentric LOFTS0050 observations from Ireland and Sweden. The station products overlap in physical frequency and integration time but have their own local noise and interference structure. BLISS was run independently on the two stations and produced 6,589 Ireland candidate rows and 5,619 Sweden candidate rows, for 12,208 raw rows in total.

7.2 Union and scoring workflow

Physical-coordinate union construction produced 12,198 entries. A total of 1,314 primary pairs followed the frozen high-resolution Stage-4 route and were scored, together with 2,848 shifted controls. Every scored primary had at least one valid control. The union is intentionally dominated by one-station detections: these are exactly the cases for which the downstream model must inspect the corresponding physical location in the other station instead of discarding the candidate.

7.3 Same-observation shifted controls

The real pilot has no truth labels for the BLISS-selected candidates, so the pipeline also scores candidate-excluded, distant-frequency control locations from the same observations. No simulated or injected signal is added to these controls. They provide a local reference for the scores produced by the real Ireland–Sweden noise and interference environment and are processed with the same frozen scoring rules as the observed candidate locations.

For primary candidate i and scoring method m , the background-relative contrast is

$$\Delta s_{i,m} = s_{i,m}^{\text{observed}} - \frac{1}{|\mathcal{C}_i|} \sum_{j \in \mathcal{C}_i} s_{ij,m}^{\text{control}}, \quad (4)$$

where $\mathcal{C}_i$ is the set of valid shifted controls associated with candidate i . Positive values indicate that the BLISS-selected location scored more match-like than its local controls; values near zero indicate no systematic score uplift relative to those controls.

Across the pilot, the Stage-4 distribution of Δs was centred close to zero: high scores occurred at observed locations, but comparably high scores also occurred at control locations. The full distributions and block-bootstrap analysis are retained in the repository. This provides an empirical background-relative characterisation of Stage-4 score behaviour on real LOFTS0050 data, while the labelled Test-A evaluation provides the quantitative discrimination measurement.

7.4 Real candidate rejection example

A fixed Sweden-only BLISS candidate provides a concrete example of how the coincidence decision changes on real data. Because the pipeline uses the catalogue union, the Sweden detection was projected into the corresponding Ireland location and both station views were compared. Under the inherited Stage-3 ranking, the candidate was near the 99th percentile of the real-pilot candidate-score distribution. After candidate-conditioned preparation and Stage-4 scoring, it moved to approximately the 1st percentile.

In operational terms, Stage 4 strongly rejected this candidate as a cross-station coincidence candidate: once the candidate was aligned and compared in physical coordinates, the Ireland–Sweden evidence was ranked as strongly mismatch-like. The value of this example is operational rather than statistical—it shows how the Stage-3-to-Stage-4 change manifests for an individual real candidate, while the labelled Test-A evaluation provides the quantitative performance result.

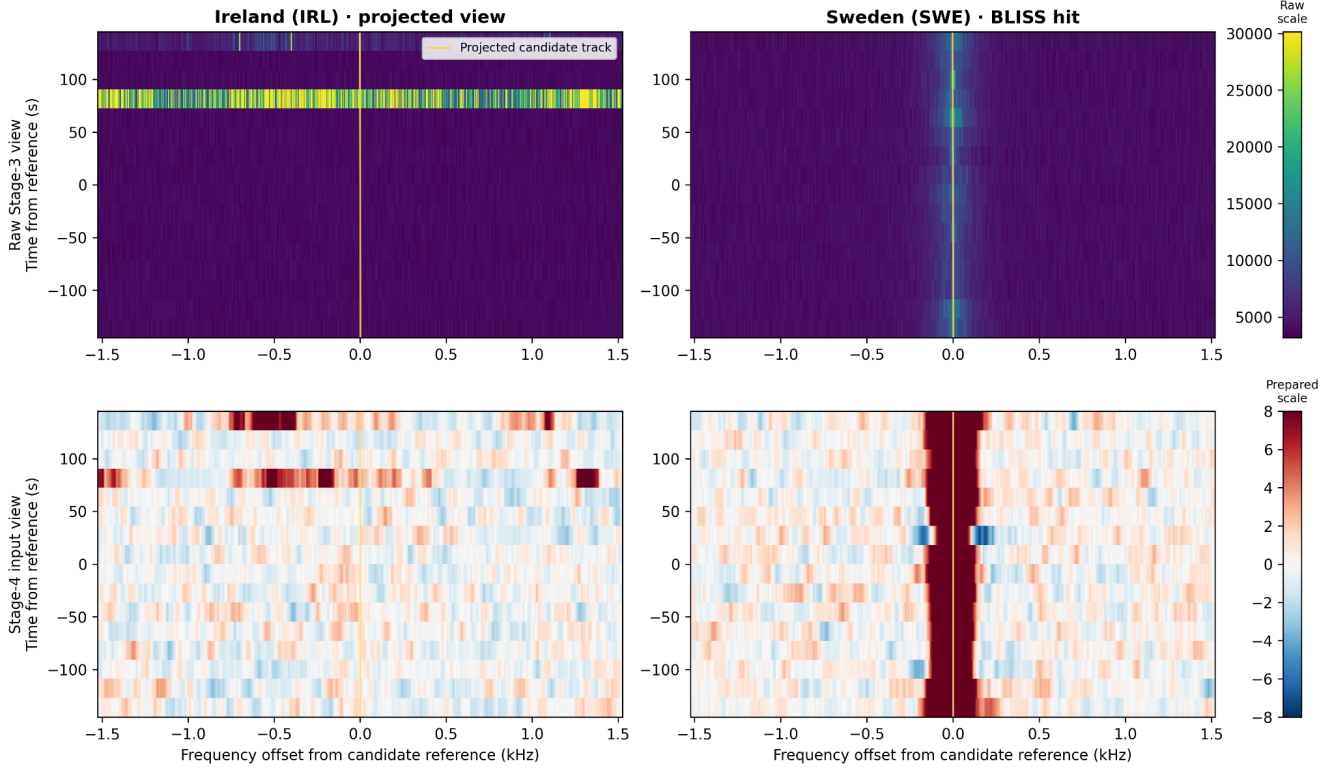


Figure 5: Real LOFTS0050 Sweden-only BLISS candidate. The upper panels show the raw Stage-3 station views; the lower panels show the corresponding candidate-conditioned Stage-4 views after drift alignment, recentering, and width-aware preparation.

8 Blind BLISS recovery of a broadened injection on real backgrounds

8.1 Purpose and setup

The blind-injection recovery validation tested whether a known broadened event can be placed into both real station backgrounds and then recovered by independent BLISS searches without exposing the truth record to the search. One Lorentzian event with FWHM 30.0 Hz, reference frequency 140.300000 MHz, signed drift $+0.0600 \text{ Hz s}^{-1}$, and requested injector strength 30.0 was added to separate copies of the Ireland and Sweden barycentric LOFTS0050 backgrounds. The original observations were not modified.

The pinned BLISS searches used the CPU/NumPy path, 262,144 fine channels per coarse channel, drift range $[-0.2, +0.2] \text{ Hz s}^{-1}$, search floor 20, and width bank $\{1, 3, 8, 20, 50, 120\}$ channels. Both station searches completed all 103 coarse channels. The recovery gate $|\Delta f| \leq 0.3 \text{ kHz}$ and $|\Delta \dot{f}| \leq 0.007 \text{ Hz s}^{-1}$ was applied only after the searches as a truth-audit rule; it was not supplied to BLISS.

8.2 Recovered candidates

Exactly one candidate fell inside the post-search recovery gate at each station.

The paired waterfall audit shows the same injected track in both station copies and the independently recovered BLISS overlays. This establishes a working paired injection–search–recovery chain on real station backgrounds:

Station	Recovered frequency (MHz)	Recovered drift (Hz s^{-1})	BLISS BANK_SNR
Ireland	140.299999	+0.05921	266.913
Sweden	140.300000	+0.05907	178.229

Table 5: Blind BLISS recovery of the injected 30 Hz broadened event. Both recovered drift estimates differ from the injected value by about $10^{-3} \text{ Hz s}^{-1}$, consistent with the nearest allowed drift-grid quantisation.

paired injection $\rightarrow$ independent BLISS search $\rightarrow$ recovery association.

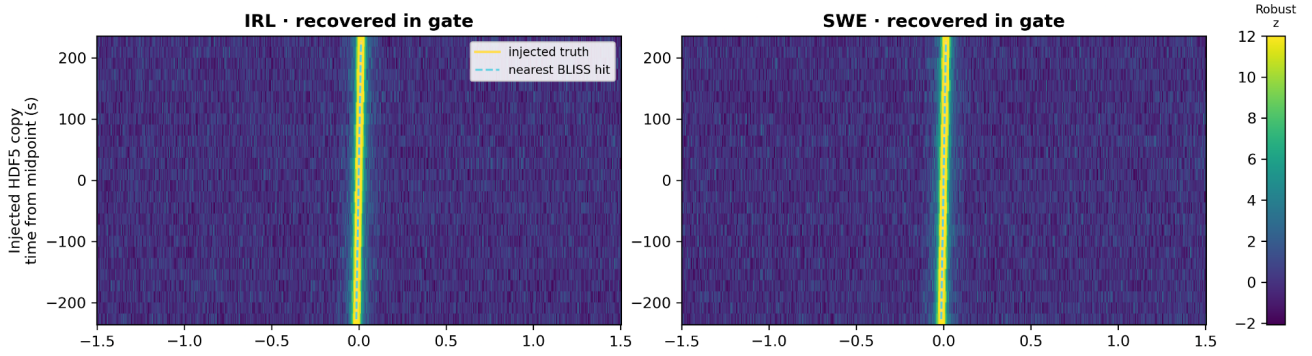


Figure 6: Blind BLISS recovery of the injected broadened event on separate copies of the real Ireland and Sweden backgrounds. The yellow overlay marks injected truth and the cyan dashed overlay is the independently recovered BLISS track. The agreement provides a direct visual check of the station-specific frequency, epoch, and signed-drift conventions.

9 Interpretation of scores and terminology

The following conventions are used throughout the evaluation and interpretation of the results:

- A **match** is a pair labelled or constructed to represent the same event at both stations. A **mismatch** is a noncoincident pair.
- A higher evaluation score is treated as more match-like. Raw/Corrected Stage 3 derive this ordering from encoder distance; Stage 4 uses the learned pair score.
- **Rejection** in this document means rejection as a two-station coincidence candidate - i.e. the pair is ranked or classified as mismatch-like. It is not a claim about the physical origin of the emission.
- The controlled Test-A population is detected-conditioned: it measures downstream coincidence discrimination once candidate information is available. BLISS search completeness is a separate upstream quantity.
- The real pilot is unlabelled and is used to characterise operational score behaviour, background-relative ranking, and candidate rejection in the coincidence workflow. Stage-4 scores are ranking scores rather than calibrated astrophysical probabilities.

10 Next research phase: multi-event Test B

The next quantitative phase of this research is to scale the validated paired injection–search–recovery chain into a locked multi-event evaluation:

1. create multiple collision-free paired injection blocks on real station backgrounds, including shared positives and deliberate noncoincident negative families;
2. keep truth segregated from BLISS search and Stage-4 inference;
3. use disjoint calibration and evaluation seeds and frequency regions;
4. run BLISS independently at Ireland and Sweden with the pinned search configuration;
5. measure BLISS recovery errors on calibration injections and freeze the recovery-association policy before evaluation;
6. form the physical-coordinate candidate union, retaining one-station detections;
7. run Stage 4 truth-blind on the recovered union;
8. write predictions before revealing the hidden labels; and
9. report BLISS completeness, Stage-4 coincidence performance conditional on candidate availability, and overall end-to-end performance as separate quantities.

The evaluation should use multiple independent blocks (currently at least five) so that uncertainty is not dominated by a single frequency region. The one-event result in Section 8 already validates the paired injection, independent BLISS search, and recovery-association chain; Test B extends that working chain into a quantitative end-to-end study. Beyond Test B, the same framework can be evaluated across additional paired backgrounds and observing epochs and across a broader range of signal and RFI conditions. The combined Stage-3-to-Stage-4 development, Test A, real-data analysis, and completed end-to-end evaluation provide a path toward a publication manuscript.

11 Conclusion

The central contribution of this internship is a candidate-conditioned dual-site coincidence pipeline for broadened BLISS candidates. The work preserves BLISS as the independent detector while making the downstream Ireland–Sweden comparison explicitly aware of candidate frequency, signed drift, centre, and spectral width. The inherited Stage-3 encoder is reused rather than discarded, but the comparison is strengthened by a physics-informed representation and a symmetric learned pair head.

On 10,800 simulated matched and mismatched candidate pairs spanning 10–100 Hz, Stage 4 increases AUC_{ROC} from 0.818 to 0.993 and true-match recovery from 44.7% to 96.3% while maintaining a false-positive rate near 3%. The ablation study shows that candidate-informed preparation provides most of the gain, with the learned Stage-4 head adding a further resolved improvement. The real LOFTS0050 pilot demonstrates the candidate union, paired extraction, scoring, controls, and rejection workflow on simultaneous barycentric data. The paired blind-injection validation then shows that BLISS independently recovers a known 30 Hz broadened event at both stations in the real background geometry.

The repository therefore captures a complete downstream Stage-4 coincidence method, an audited real-data workflow, and a validated paired injection–recovery chain. The next phase of this research is the multi-event Test B, followed by broader validation across additional paired backgrounds and signal/RFI conditions. Together, these results provide the basis for a publication-level evaluation of broadened dual-site coincidence scoring.